

Rachel Hopper

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SUMMARY:

Highly motivated Physics student with strong technical proficiency in programming (C++, Python, Java) and engineering design (Siemens NX CAD). Experienced in applying theoretical knowledge to practical aerospace, robotics, and defense challenges through NASA-affiliated projects and a NORAD & USNORTHCOM internship. Skilled in collaborative problem solving, embedded systems, mechanical design, and technical presentations. Cleared for work in sensitive environments and eager to leverage a strong foundation in physics and engineering to contribute to innovative scientific, defense, and technical endeavors.

EDUCATION:

University of Colorado at Colorado Springs 05/2026

Bachelor of Science in Physics, GPA: 3.59/4.00

Relevant Courses: Math Methods Phys-Engr, Electricity & Magnetism, Quantum Mechanics, Optics

SKILLS:

Computer Languages: Java, C, C++, Python, R

Operating Systems: Windows, Linux

Software: Siemens NX CAD

Data Analysis: Excel, AI, R, Power BI

WORK EXPERIENCE

NORAD and USNORTHCOM Volunteer Student Internship

01/2026-03/2026

- Obtained security clearance and participated in classified activities.
- Supported daily procedures and briefings with homeland security teams.
- Gained exposure to military operations, defense strategy, and interagency coordination.

PROJECT WORK EXPERIENCE:

NPWEE, NASA L'Space Program

09/2025-12/2025

Overview: Designed to learn the proposal writing process, gain experience evaluating proposals, and collaborate with a team to submit a proposal.

- Innovate to help address NASA pain points
- Dedicate time for collaboration, research, writing, and submitting all team and individual deliverables
- Participating in a NASA Proposal Review Panel

Mission Concept Academy, NASA L'Space Program

01/2023 -05/2023

Overview: In a team of 6, developed a mission for a rover to study the Mars Northern Polar deposits.

- Co-led a team of six to design a mission concept targeting the analysis of Martian northern polar deposits.
- Designed a Mars rover model using Siemens NX CAD, focusing on component integration and terrain adaptability.
- Authored the mission proposal and presented design justifications and scientific goals to an expert evaluation panel.

Rocketry, University of Alabama

08/2021 - 05/2022

Overview: Competed in a NASA competition to create a rocket to drop a payload and give its location without using GPS.

- Developed C++ software for a custom payload location system without using GPS, applying embedded systems logic and sensor integration
- Collaborated in the systems engineering process to design, test, and simulate rocket deployment and payload recovery
- Contributed to the technical documentation and co-authored the final project presentation submitted to NASA